

Track D: Finding errors in code

Generated from the same content files as the app. Each module lists its roadmap months. Solutions are in the appendix.

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Track D: Finding errors in code

Find and fix errors fast, including errors in code that Claude and Copilot write.

D3 Silent errors

Roadmap: Every month. Join explosion, nulls in filters, type coercion, timezone drift, mutable defaults.

D3-L3 Mutable defaults and the truthiness of zero

Concept

Two Python habits produce wrong results with no error message. Both appear in code written by people and by AI assistants, including your reference pipeline.

Mutable default arguments. Python evaluates a default value once, when it defines the function. Every call that relies on the default shares that one object. Write `def add_check(result, results=[])`, and each call appends to the same list. The second validation run returns the first run's checks as well.

The fix is a `None` sentinel:

```
1 def add_check(result: str, results: list[str] | None = None) -> list[str]:
2     results = [] if results is None else results
3     results.append(result)
4     return results
```

ruff flags the pattern as rule B006. Treat that warning as a bug.

Truthiness of numbers. `if value:` is `False` for `None`, and also for `0`, `0.0`, empty strings and empty containers. When zero is a valid measurement, such as a 0% reach, a zero weight or zero rows, a truthiness test treats it as missing.

Your reference `check_wave_stability` does this:

```
1 if current_pct and prior_pct:
```

A metric that falls from 12% to 0% skips the comparison. The largest possible drop produces no warning.

Test for the case you mean. For missing, write `is None`. For zero, write `== 0`.

```
1 if current_pct is not None and prior_pct is not None:
```

The same rule covers type hints. `error_message: str = None` claims the value is always a string. Write `str | None = None`, and mypy then makes you handle the `None` case.

Why it matters

A questionnaire routing error removes one question for a whole wave. Its weighted percentage drops from 12% to 0.0%. The wave-over-wave stability check runs `if current_pct and prior_pct:`, treats 0.0 as missing and skips the metric. QA reports PASS. The dashboard publishes 0% reach for that question, and a client asks why the channel "lost its whole audience" in a week.

The check existed to catch exactly this change. One truthiness test disabled it for the one value that mattered most.

Worked example: weak code

```
❷ def compare_waves(current: dict, prior: dict, tolerance: float = 0.10,
    flags=[]):
    2     """Flag metrics whose share moved more than tolerance between
        waves."""
    3     for metric, current_pct in current.items():
    4         prior_pct = prior.get(metric)
    ❸     if current_pct and prior_pct:
    6         if abs(current_pct - prior_pct) > tolerance:
    7             flags.append(metric)
    8     return flags
```

1. **Line 1.** Bare `dict` hides the value type. The values are floats or `None`.
2. **Line 1.** The default list is created once. Flags from one call leak into the next.
3. **Line 5.** A share of 0.0 is falsy, so the biggest possible drop is never compared.

Worked example: strong code

```
1 def compare_waves(
    ❶     current: dict[str, float | None],
    3     prior: dict[str, float | None],
    4     tolerance: float = 0.10,
    5 ) -> list[str]:
    6     """Flag metrics whose share moved more than tolerance between waves.
    7
    8     A metric is skipped only when a share is missing. A share of 0.0 is
    9     compared.
    10    """
    ❷     flags: list[str] = []
    11     for metric, current_pct in current.items():
    12         prior_pct = prior.get(metric)
    ❸     if current_pct is None or prior_pct is None:
    14         continue
    15         if abs(current_pct - prior_pct) > tolerance:
    16             flags.append(metric)
    17     return flags
```

1. **Line 2.** The hint says a share can be missing, so mypy checks every use.

- 2. **Line 10.** A new list on every call. No state leaks between runs.
- 3. **Line 13.** Skips only missing values. 0.0 goes on to the comparison.

Both fixes are one line. Both bugs pass every test that uses only non-zero values and a single call.

Recap card

- **When does Python evaluate a default argument?** Once, when it defines the function. A list or dict default is shared by every call.
- **Safe pattern for an optional list argument** Default to None. Inside: `results = [] if results is None else results.`
- **What is wrong with** `if current_pct and prior_pct: ?` 0.0 is falsy, so a real share of zero is treated as missing. Use `is not None`.
- **Type hint for an optional string that defaults to None** `str | None = None`. Never `str = None`.

Practice

Exercise 1. Predict the output (level 1)

What does this print?

```
1 def add_flag(flag, flags=[]):
2     flags.append(flag)
3     return flags
4
5 add_flag("null_weights")
6 print(add_flag("row_count"))
```

Exercise 2. Multiple choice (level 2)

`current_pct = 0.0` and `prior_pct = 0.12`. The tolerance is 0.10. What does `check_wave_stability` from the reference code do with this metric?

- a) Flags it as WARN, because the change is 0.12.
- b) Marks it PASS.
- c) Leaves it out of the results.
- d) Raises a ZeroDivisionError.

Why?

- r1) `if current_pct and prior_pct:` is False when `current_pct` is 0.0, so the block never runs.
- r2) The function filters out metrics below 1%.

Exercise 3. Fill the gap (level 3)

Fix both signatures. The error message is optional, and the flags list must not be shared between calls.

```
1 def log_run(status: str, error_message: ____ (1) = None) -> None: ...
2
3 def collect(flag: str, flags: list[str] | None = ____ (2)) ->
  list[str]: ...
```

Exercise 4. Spot the bug (level 3)

This check should warn when a share moves by more than the tolerance. A drop from 12% to 0% produces no warning. Tap the line with the bug, then fix the function.

```
1 def stability_status(current_pct: float | None, prior_pct: float | None,
  tolerance: float = 0.10) -> str:
2     """Return WARN, PASS, or SKIP when a share is missing."""
3     if current_pct and prior_pct:
4         change = abs(current_pct - prior_pct)
5         return "WARN" if change > tolerance else "PASS"
6     return "SKIP"
```

Write down the line number of the bug. Then write the fix.

Exercise 5. Write the function (level 4)

Write `compare_waves(current, prior, tolerance=0.10)`. Both arguments map metric names to a share or `None`. Return the metrics whose share changed by more than `tolerance`, in the order of `current`. Skip a metric only when either share is `None` or the metric is missing from `prior`. Calling the function twice must not carry results over.

Starter code:

```
1 def compare_waves(
2     current: dict[str, float | None],
3     prior: dict[str, float | None],
4     tolerance: float = 0.10,
5 ) -> list[str]:
6     ...
```

Exercise 6. Refactor (level 5)

A validator collects check results with the helper below. The second contract validation in a job reports the first contract's failures too. Refactor `add_result` so each validation starts empty, keep the signature compatible for callers that pass a list, and add type hints and a docstring.

Starter code:

```
1 def add_result(name, passed, results=[]):  
2     results.append({"check": name, "passed": passed})  
3     return results
```

Sources

- ruff rule B006 flags mutable default arguments. (verified: ruff 0.16.9 run in this repo reports B006 on `def f(x=[])`.)

Track D best-practice checklist

- ☐ Read a traceback from the bottom up. Find the first frame in your own code.
- ☐ Reproduce the error on the smallest data that shows it.
- ☐ Check row counts before and after every join.
- ☐ Test numbers with `is None`, never with truthiness, when zero is a valid value.
- ☐ Never use a mutable object as a default argument.
- ☐ Check every function and parameter an AI suggests against the official docs for your version.
- ☐ Run AI code on a small input with a known answer before you trust it.
- ☐ Write the failing test first. A test that only repeats the code proves nothing.
- ☐ Ask the model to list its assumptions. Check each one.
- ☐ State versions, constraints, expected outputs and tests in every prompt.

Appendix: solutions and answer keys

Attempt each exercise before you read its solution. Track D answer keys are here too.

D3-L3 Mutable defaults and the truthiness of zero

D3-L3 exercise 1

Hints. (1) When is the default `[]` created? (2) Both calls use the default. Is it the same list? (3) The list is created once, when Python defines the function.

Answer.

```
1 ['null_weights', 'row_count']
```

The default list is created when Python defines the function. Both calls append to that one list, so the second call returns both flags.

Common wrong answers.

- `['row_count']`: That is what the author intended. The default list is created once, so the second call appends to the list the first call used.
- `['row_count', 'null_weights']`: `append` adds to the end. The first call's flag is already in the list.

D3-L3 exercise 2

Hints. (1) Evaluate `0.0` and `0.12` in your head. (2) `0.0` is falsy, like `None`. (3) The whole block, including `results.append`, is skipped.

Answer. c) Leaves it out of the results. Because: r1) `if current_pct and prior_pct:` is False when `current_pct` is 0.0, so the block never runs.

`0.0` and `0.12` evaluates to `0.0`, which is falsy. The block that compares and appends never runs, so the metric vanishes from the results and no warning appears.

Common wrong answers.

- a: That is what the check should do. Look at the `if` line before the comparison: `0.0` is falsy.
- b: Nothing is appended for this metric, so it cannot be marked PASS. The `if` skips it.
- d: The check subtracts, it does not divide. No error appears, which is the problem.

- `c|r2`: Right outcome, wrong reason. There is no 1% filter. The truthiness test treats 0.0 as missing.

D3-L3 exercise 3

Hints. (1) The first default is `None`, so the type must allow `None`. (2) The second default must be immutable. (3) `str | None` for the first gap and `None` for the second.

Answer.

```
1 def log_run(status: str, error_message: str | None = None) -> None: ...
2
3 def collect(flag: str, flags: list[str] | None = None) -> list[str]: ...
```

`error_message: str | None = None` states that the value may be missing.

`flags: list[str] | None = None` avoids a shared list. Create the list inside the function when it is `None`.

Common wrong answers.

- `g1 = str: str = None` claims the value is always a string. mypy in strict mode rejects it. Write `str | None`.
- `g2 = []`: A list default is created once and shared by every call. Use `None` and create the list inside the function.

D3-L3 exercise 4

Hints. (1) Which values make `if current_pct and prior_pct`: `False`? (2) 0.0 is one of them. (3) Replace the truthiness test with `is not None` on both values.

Answer. The bug is on line 3.

Fixed version:

```
1 def stability_status(current_pct: float | None, prior_pct: float | None,
2                       tolerance: float = 0.10) -> str:
3     """Return WARN, PASS, or SKIP when a share is missing."""
4     if current_pct is not None and prior_pct is not None:
5         change = abs(current_pct - prior_pct)
6         return "WARN" if change > tolerance else "PASS"
7     return "SKIP"
```

Line 3 treats 0.0 as missing. Use `if current_pct is not None and prior_pct is not None` so that only missing shares skip the comparison.

Common wrong answers.

- Line 4: The subtraction is correct. The code never reaches it when a share is 0.0.

- Line 5: The comparison with the tolerance is correct. Look at what stops the code reaching it.

D3-L3 exercise 5

Hints. (1) Start with `flags: list[str] = []` inside the function. (2) Read the prior share with `prior.get(metric)`, which gives `None` for a missing metric. (3) Skip with `if current_pct is None or prior_pct is None: continue`, then compare with the tolerance.

Answer.

```

1 def compare_waves(
2     current: dict[str, float | None],
3     prior: dict[str, float | None],
4     tolerance: float = 0.10,
5 ) -> list[str]:
6     """Flag metrics whose share moved more than tolerance between waves.
7
8     A metric is skipped only when a share is missing. A share of 0.0 is
    compared.
9     """
10    flags: list[str] = []
11    for metric, current_pct in current.items():
12        prior_pct = prior.get(metric)
13        if current_pct is None or prior_pct is None:
14            continue
15        if abs(current_pct - prior_pct) > tolerance:
16            flags.append(metric)
17    return flags

```

Create the list inside the function. Use `prior.get` so a new metric gives `None`. Skip only on `None`, then compare the absolute change with the tolerance.

D3-L3 exercise 6

Hints. (1) Default `results` to `None`. (2) Create a new list only when `results` is `None`. (3) `results or []` looks similar but replaces an empty list the caller passed. Use `is None`.

Answer.

```

1 def add_result(
2     name: str, passed: bool, results: list[dict[str, object]] | None =
    None
3 ) -> list[dict[str, object]]:
4     """Append one check result and return the list.
5
6     Args:
7         name: Check name.
8         passed: Whether the check passed.
9         results: List to append to. A new list is created when omitted.

```

```
10
11     Returns:
12         The list with the new result.
13     """
14     results = [] if results is None else results
15     results.append({"check": name, "passed": passed})
16     return results
```

Default to `None`, create the list when `results is None`, then append and return. Avoid `results or []`: an empty list the caller passed is falsy, so the function would append to a new list and the caller would never see the result.